

moment about the lateral axis is studied in the coefficient form.

$$M = C_M q S (MAC)$$

or

$$C_M = \frac{M}{q S (MAC)}$$

where

M = pitching moment about the c.g., ft.-lbs., positive if in a nose-up direction

q = dynamic pressure, psf

S = wing area, sq. ft.

MAC = mean aerodynamic chord, ft.

C_M = pitching moment coefficient

The pitching moment coefficients contributed by all the various components of the aircraft are summed up and plotted versus lift coefficient. Study of this plot of C_M versus C_L will relate the static longitudinal stability of the airplane.

Graph *A* of figure 4.5 illustrates the variation of pitching moment coefficient, C_M , with lift coefficient, C_L , for an airplane with positive static longitudinal stability. Evidence of static stability is shown by the tendency to return to equilibrium—or “trim”—upon displacement. The airplane described by graph *A* is in trim or equilibrium when $C_M = 0$ and, if the airplane is disturbed to some different C_L , the pitching moment change tends to return the aircraft to the point of trim. If the airplane were disturbed to some higher C_L (point *Y*), a negative or nose-down pitching moment is developed which tends to decrease angle of attack back to the trim point. If the airplane were disturbed to some lower C_L (point *X*), a positive, or nose-up pitching moment is developed which tends to increase the angle of attack back to the trim point. Thus, positive static longitudinal stability is indicated by a negative slope of C_M versus C_L , i.e., positive stability is evidenced by a decrease in C_M with an increase in C_L .

The degree of static longitudinal stability is indicated by the slope of the curve of pitching moment coefficient with lift coefficient. Graph

B of figure 4.5 provides comparison of the stable and unstable conditions. Positive stability is indicated by the curve with negative slope. Neutral static stability would be the result if the curve had zero slope. If neutral stability exists, the airplane could be disturbed to some higher or lower lift coefficient without change in pitching moment coefficient. Such a condition would indicate that the airplane would have no tendency to return to some original equilibrium and would not hold trim. An airplane which demonstrates a positive slope of the C_M versus C_L curve would be unstable. If the unstable airplane were subject to any disturbance from equilibrium at the trim point, the changes in pitching moment would only magnify the disturbance. When the unstable airplane is disturbed to some higher C_L , a positive change in C_M occurs which would illustrate a tendency for continued, greater displacement. When the unstable airplane is disturbed to some lower C_L , a negative change in C_M takes place which tends to create continued displacement.

Ordinarily, the static longitudinal stability of a conventional airplane configuration does not vary with lift coefficient. In other words, the slope of C_M versus C_L does not change with C_L . However, if the airplane has sweepback, large contribution of power effects to stability, or significant changes in downwash at the horizontal tail, noticeable changes in static stability can occur at high lift coefficients. This condition is illustrated by graph *C* of figure 4.5. The curve of C_M versus C_L of this illustration shows a good stable slope at low values of C_L . Increasing C_L effects a slight decrease in the negative slope hence a decrease in stability occurs. With continued increase in C_L , the slope becomes zero and neutral stability exists. Eventually, the slope becomes positive and the airplane becomes unstable or “pitch-up” results. Thus, at any lift coefficient, the static stability of the airplane is depicted by the slope of the curve of C_M versus C_L .